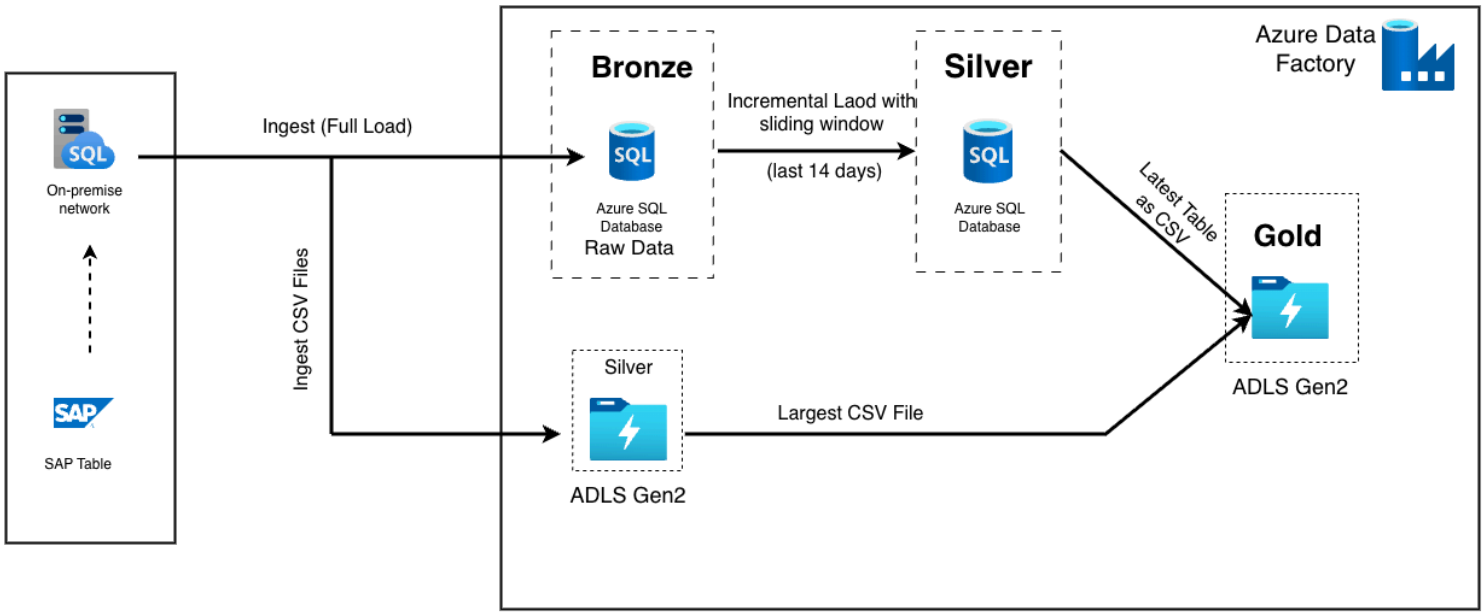
Project Name : SAP Data Pipeline - Medallion Architecture

Tools Used: Azure Data Factory, SQL Server, Azure SQL Database, Azure Data Lake Storage Gen2

Project Overview

This project implements an end-to-end data pipeline using Azure Data Factory to replicate 6 SAP tables (MARA, VBAP, VBRK, COSTLIST, INVOICE_LINE_ITEM, PRODUCING_STAT_RATIO_DETAIL) from On-Premise SQL Server into a Medallion Architecture (Bronze/Silver/Gold) on Azure, enabling incremental data loads with a sliding 14-day window for reporting purposes. Three tables are loaded as CSV files into the target silver ADLS Gen2 and only the largest file is copied into the target Gold ADLS Gen2, whereas three tables are loaded as SQL tables into bronze schema in Azure SQL Source DB and incremental load of 14 days data is loaded into silver schema in Azure SQL Target DB and the latest table is copied to the target Gold Data Lake Storage as CSV file.

Architecture Overview



SQL Server (Source)
→ Raw SAP table replicas, 3 tables

Bronze Layer (Azure SQL)
→ Full load every run, raw copy of source
→ Delete and reload each pipeline run

Silver Layer (Azure SQL)
→ Incremental load with watermark table
→ Sliding 14 day window maintained
→ Upsert with key column per table

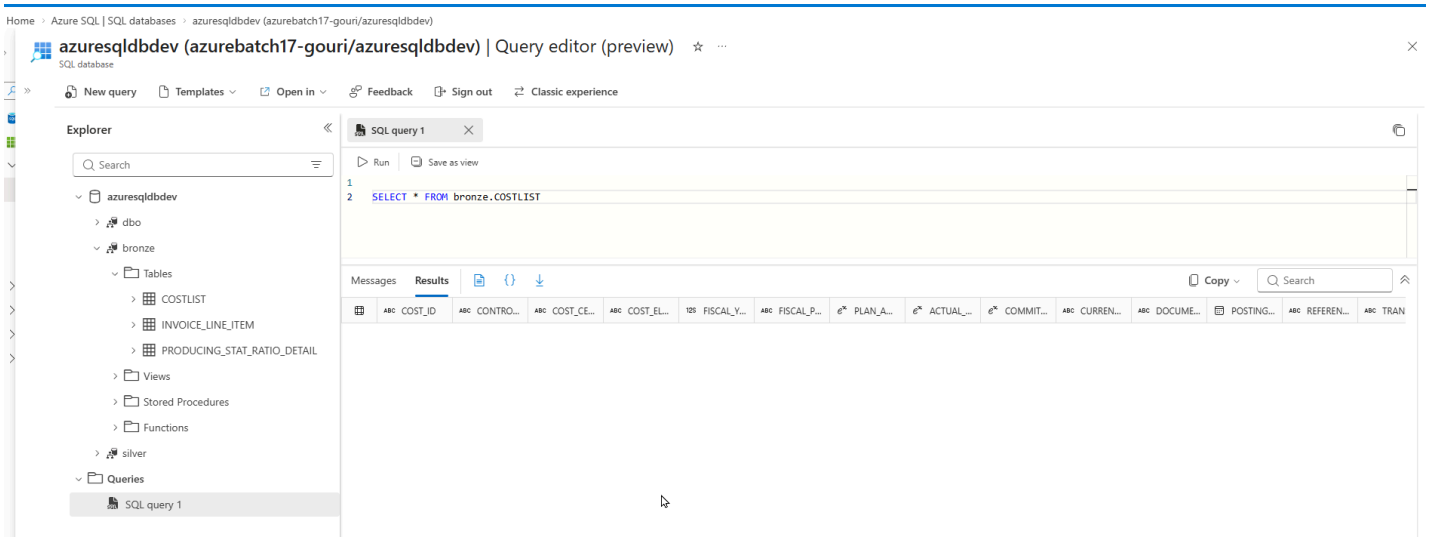
Silver Layer (ADLS Gen2)
→ CSV copy of all files

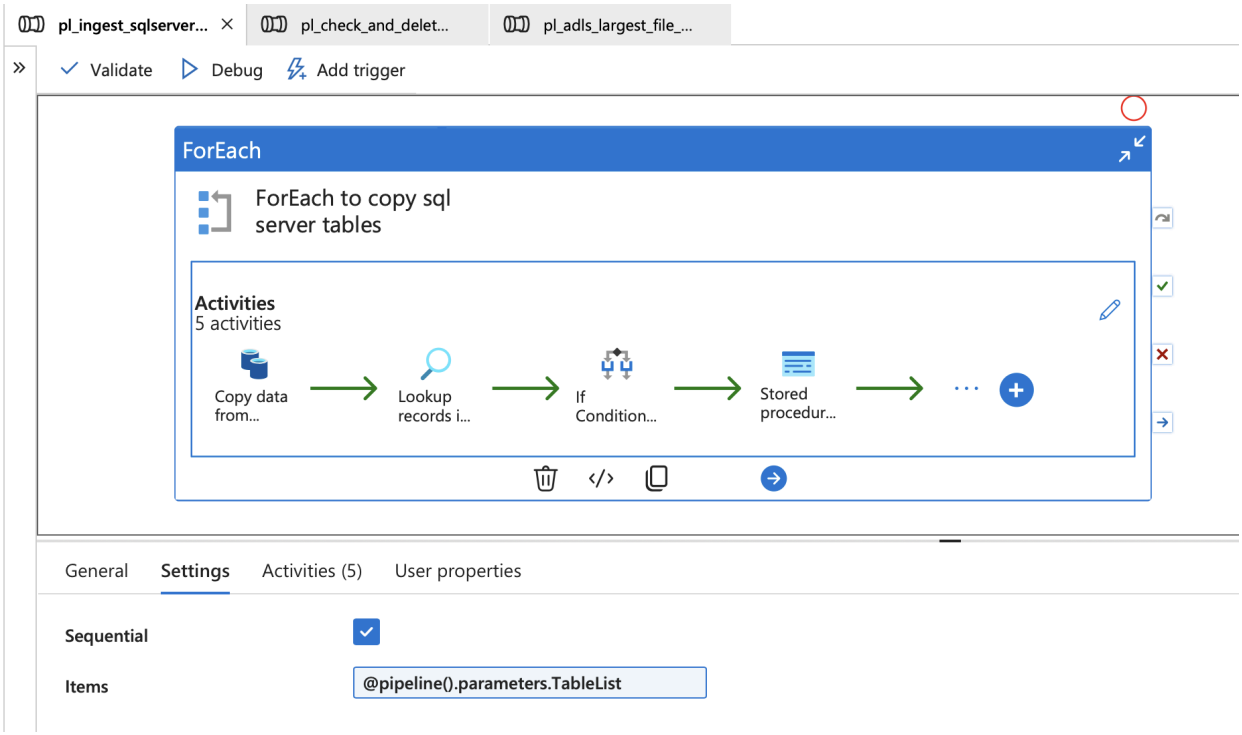
Gold Layer (ADLS Gen2)
→ CSV export of latest silver table from SQL
→ Largest CSV export from ADLS silver layer
→ Final consumption layer for reporting

Pipeline Design

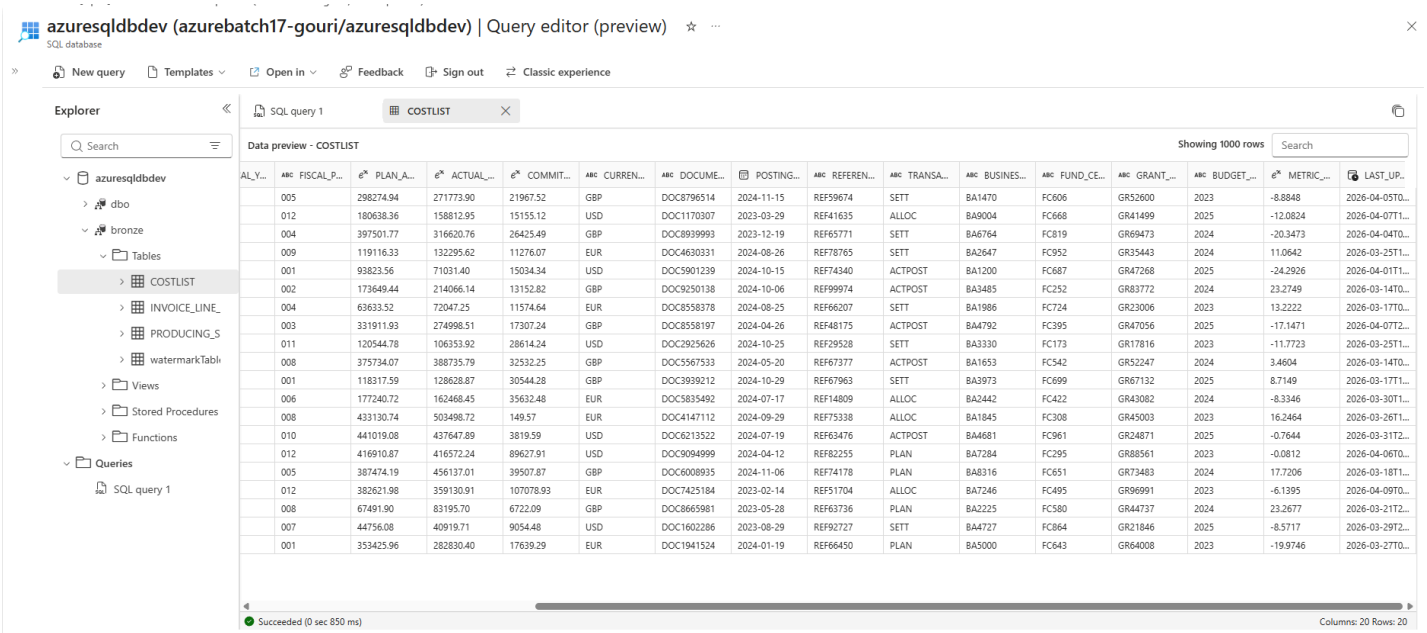
1. Pl_ingest_sqlserver_to_bronze

SQL Source Bronze layer:





Copy tables from on-premise SQL server to Raw or Bronze layer

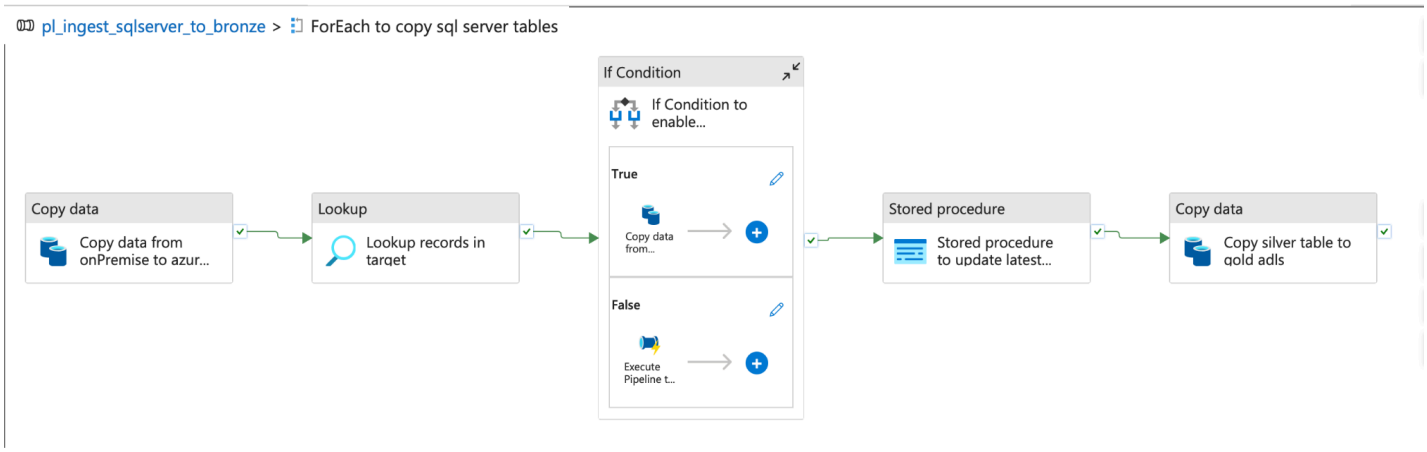


AL_Y...	FISCAL_P...	PLAN_A...	ACTUAL_...	COMMIT...	CURREN...	DOCUME...	POSTING...	REFEREN...	TRANSA...	BUSINES...	FUND_CE...	GRANT_...	BUDGET_...	METRIC_...	LAST_UP...
	005	298274.94	271773.90	21967.52	GBP	DOC8796514	2024-11-15	REF59674	SETT	BA1470	FC606	GR52600	2023	-8.8848	2026-04-05T0...
	012	180638.36	158812.95	15155.12	USD	DOC1170307	2023-03-29	REF41635	ALLOC	BA9004	FC668	GR41499	2025	-12.0824	2026-04-07T1...
	004	397501.77	316620.76	26425.49	GBP	DOC8939993	2023-12-19	REF65771	SETT	BA6764	FC819	GR69473	2024	-20.3473	2026-04-04T0...
	009	119116.33	132295.62	11276.07	EUR	DOC4630331	2024-08-26	REF78765	SETT	BA2647	FC952	GR35443	2024	11.0642	2026-03-25T1...
	001	93823.56	71031.40	15034.34	USD	DOC5901239	2024-10-15	REF74340	ACTPOST	BA1200	FC687	GR47268	2025	-24.2926	2026-04-01T1...
	002	173649.44	214066.14	13152.82	GBP	DOC9250138	2024-10-06	REF99974	ACTPOST	BA3485	FC252	GR83772	2024	23.2749	2026-03-14T0...
	004	63633.52	72047.25	11574.64	EUR	DOC8558378	2024-08-25	REF66207	SETT	BA1986	FC724	GR23006	2023	13.2222	2026-03-17T0...
	003	331911.93	274998.51	17307.24	GBP	DOC8558197	2024-04-26	REF48175	ACTPOST	BA4792	FC395	GR47056	2025	-17.1471	2026-04-07T2...
	011	120544.78	106353.92	28614.24	USD	DOC2925626	2024-10-25	REF29528	SETT	BA3330	FC173	GR17816	2023	-11.7723	2026-03-25T1...
	008	375734.07	388735.79	32532.25	GBP	DOC5567533	2024-05-20	REF67377	ACTPOST	BA1653	FC542	GR52247	2024	3.4604	2026-03-14T0...
	001	118317.59	128628.87	30544.28	GBP	DOC3939212	2024-10-29	REF67963	SETT	BA3973	FC699	GR67132	2025	8.7149	2026-03-17T1...
	006	177240.72	162468.45	35632.48	EUR	DOC5835492	2024-07-17	REF14809	ALLOC	BA2442	FC422	GR43082	2024	-8.3346	2026-03-30T1...
	008	433130.74	503498.72	149.57	EUR	DOC4147112	2024-09-29	REF75338	ALLOC	BA1845	FC308	GR45003	2023	16.2464	2026-03-26T1...
	010	441019.08	437647.89	3819.59	USD	DOC6213522	2024-07-19	REF63476	ACTPOST	BA4681	FC961	GR24871	2025	-0.7644	2026-03-31T2...
	012	416910.87	416572.24	89627.91	USD	DOC9094999	2024-04-12	REF82255	PLAN	BA7284	FC295	GR88561	2023	-0.0812	2026-04-06T0...
	005	387474.19	456137.01	39507.87	GBP	DOC6008935	2024-11-06	REF74178	PLAN	BA8316	FC651	GR73483	2024	17.7206	2026-03-18T1...
	012	382621.98	359130.91	107078.93	EUR	DOC7425184	2023-02-14	REF51704	ALLOC	BA7246	FC495	GR96991	2023	-6.1395	2026-04-09T0...
	008	67491.90	83195.70	6722.09	GBP	DOC8665981	2023-05-28	REF63736	PLAN	BA2225	FC580	GR44737	2024	23.2677	2026-03-21T2...
	007	44756.08	40919.71	9054.48	USD	DOC1602286	2023-08-29	REF92727	SETT	BA4727	FC864	GR21846	2025	-8.5717	2026-03-29T2...
	001	353425.96	282830.40	17639.29	EUR	DOC1941524	2024-01-19	REF66450	PLAN	BA5000	FC643	GR64008	2023	-19.9746	2026-03-27T0...

Activity 1: ForEach

Purpose: Iterates over 3 tables dynamically

Parameter passed: tableName, schemaName, timestampColumn, keyColumn

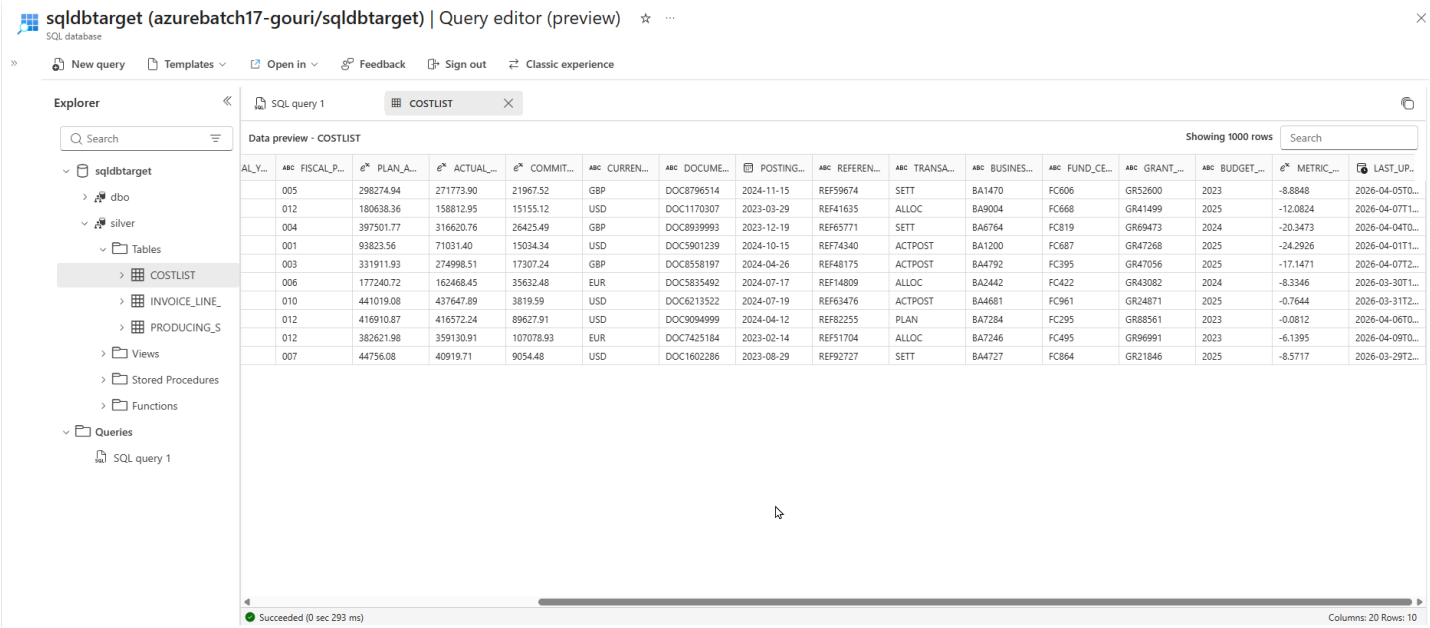


Activity 2: Copy (On-Premise SQL Server → Bronze)
Purpose: Full load, delete and reload
Pre-copy script: DELETE TABLE bronze.tableName

Activity 3: Lookup (COUNT rows in silver)
Purpose: Check if silver table is empty
Query: SELECT COUNT(*) AS row_count FROM silver.tableName

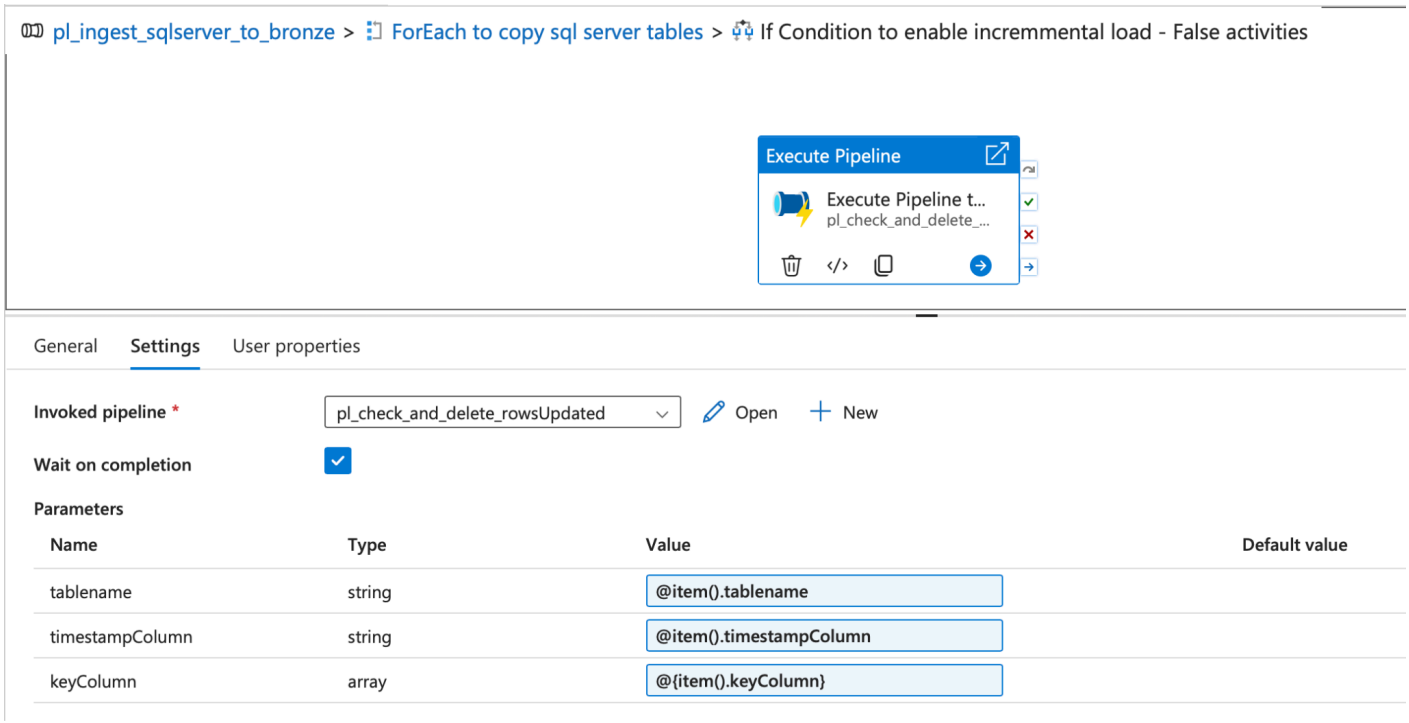
Activity 4: If Condition
Purpose: Route to initial or incremental load
Expression: @equals(activity('Lookup').output.firstRow.row_count, 0)

Copy last 14 days data from Bronze to SQL Target Silver layer



TRUE Branch
Step 1: Copy last 14 days Bronze → Silver

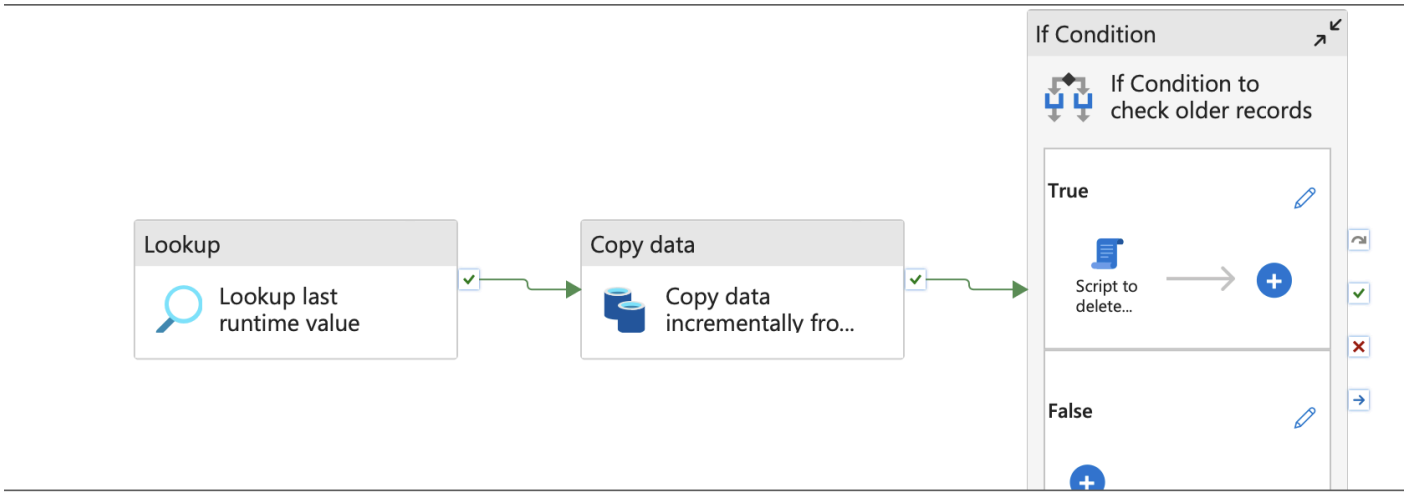
FALSE Branch
Step 1: Execute Child Pipeline
Parameters passed: Shown in the below figure.

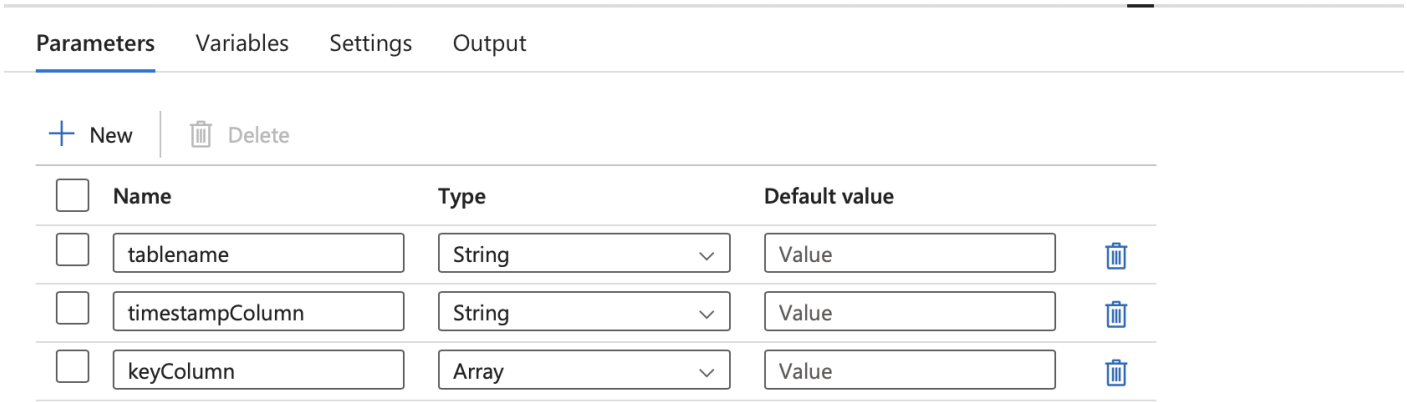


After initial copy or incremental load,
Activity 5: Stored Procedure
Purpose**:** Update the watermark table with the latest timestamp

Activity 6: Copy (Azure SQL Silver Target → ADLS Gold)
Purpose: Copy the latest updated table to the ADLS Gold layer as CSV file

2. pl_check_and_delete_rowsUpdated

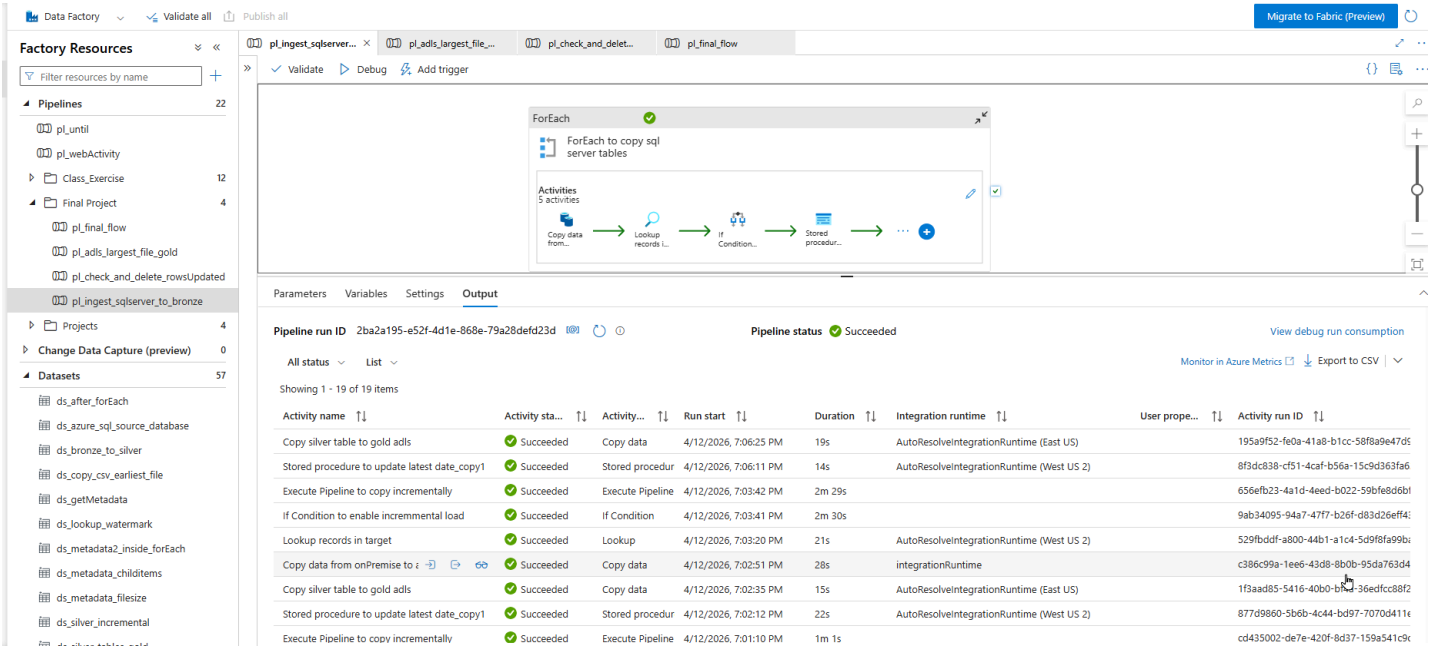




Step 1: **Lookup** - GET watermark date
Step 2: **Copy Activity** - Bronze → Silver incremental upsert
  Source query: WHERE timestampColumn > watermark AND <= utcNow()
Step 3: **If Condition** : rowsCopied > 0?

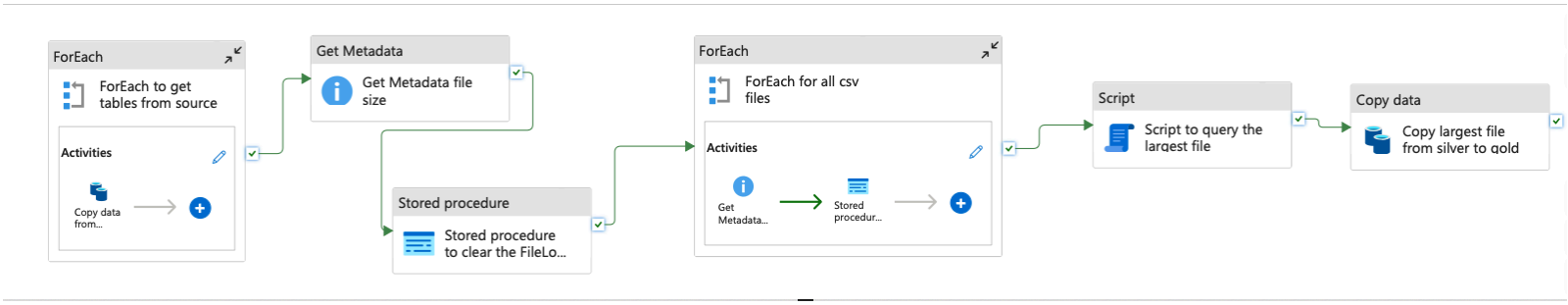
  TRUE Branch:
    Script Activity - Delete records older than 14 days
DELETE FROM silver.@{pipeline().parameters.tablename} WHERE @{pipeline().parameters.timestampColumn} < '@{formatDateTime(adddays(utcNow(), 14), 'yyyy-MM-ddTHH:mm:ss')}'

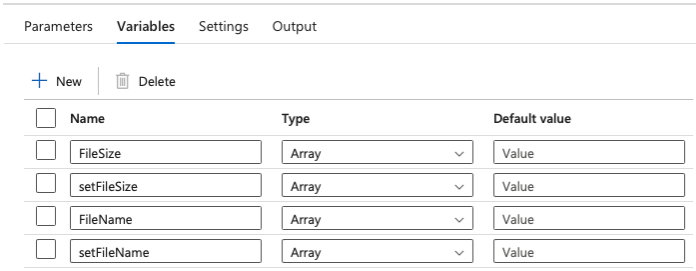
  FALSE:
    Do nothing



3. Pl_adls_largest_file_gold

Activity 1: ForEach (MARA, VBRK, VBAP)
Step 1: Copy Activity **-** Excel file in ADLS Source to Silver ADLS folder as CSV



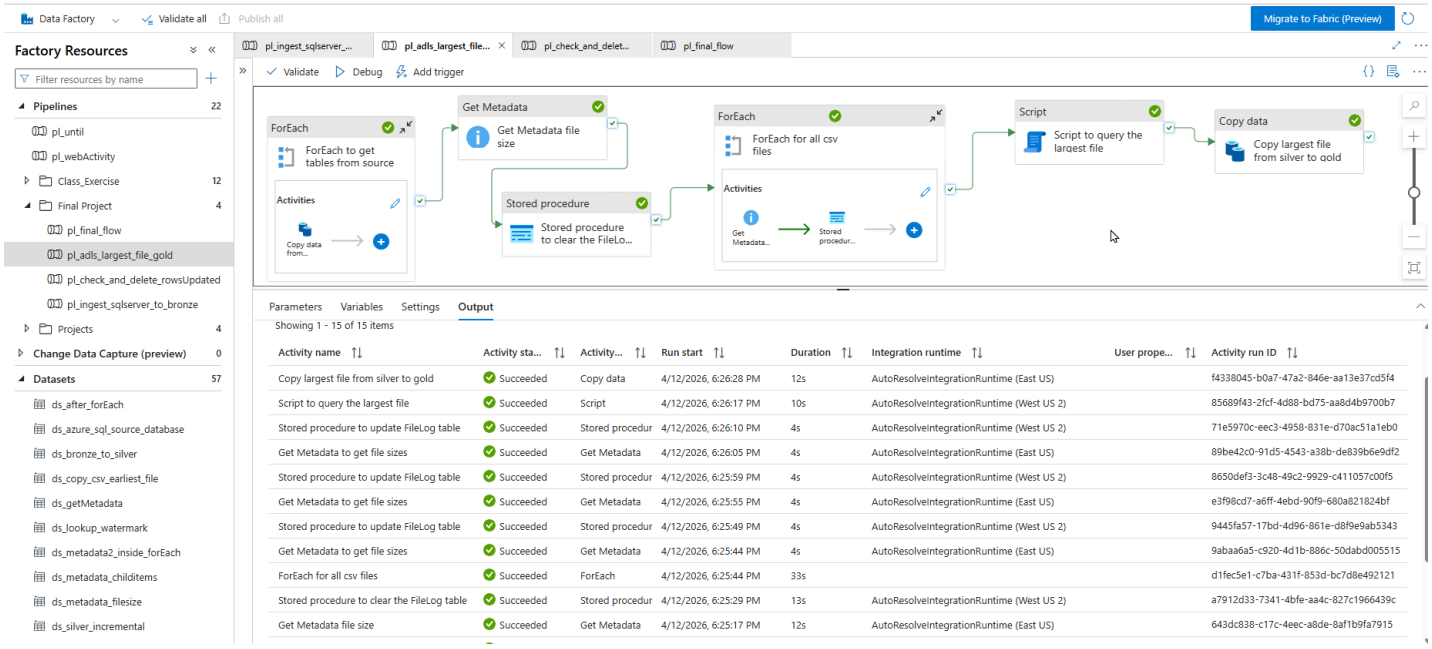


Activity 2: Get Metadata (after ForEach completes)
Purpose: Gets list of all child items (files) in silver folder

Activity 3: Stored Procedure (usp_logFileDetails**)**
Purpose: Clears the FileLog table before logging fresh file details

Activity 4: ForEach (iterates over all CSV files found in silver)
→ Get Metadata per file
→ Gets size and name of each individual file
→ Stored Procedure: usp_logFileDetails
    → Inserts filename and filesize into silver.FileLog table

Activity 5: Script Activity
Queries silver.FileLog to find filename with largest size:

```
SELECT filename FROM silver.FileLog
WHERE filesize = (SELECT MAX(filesize) FROM silver.FileLog)
```

Activity 6: **Copy Activity**
Source: Silver ADLS largest file
Sink: Gold ADLS folder
Copies only the largest file to gold